

AMENDMENT NO. 1 JANUARY 1973
TO
SP : 8 - 1970 ISI HANDBOOK OF MATHEMATICAL, PHYSICAL,
CHEMICAL AND ENGINEERING TABLES

Addenda

This amendment to the **ISI** Handbook of Mathematical, Physical, Chemical and Engineering Tables (SP : 8 - 1970) introduces two new tables, namely, **18A** and **25A** in Section 2 Physical and Chemical Tables; and three new tables, namely, **32A**, **37A** and **37B** in Section 3 Engineering Tables. Besides, four existing tables, namely, Tables 33, 34, 35 and 37, in Section 3, have been

enlarged to incorporate a column each giving data on centre of gravity, perimeter, etc. In another Table 45, the amount of data added is so substantial that the **revised** table replaces the existing Table 45 on page 162 of the Handbook.

The details of the contents of the amendment **are** as under, which should be read in conjunction with the Handbook (SP : 8 - 1970):

TABLE No.	TITLE	REMARKS
Section 2 Physical and Chemical Tables		
18A	Psychrometric Properties of Air	} These two are new tables introduced in the Handbook
25A	Some Physical Constants	
Section 3 Engineering Tables		
32A	Rolled Steel Heavy Beams	New table introduced
33	Rolled Steel Channels	Introduces data on Centre of Gravity in a new column No. 12 in Table 33 (P 154)
34	Steel Equal Angles	Introduces data on Centre of Gravity in a new column No. 11 in Table 34 (P 155)
35	Steel Tee Bars	Introduces data on Centre of Gravity Position in a new column No. 10 in Table 35 (P 157)
37	Round and Square Steel Bars	Introduces data on Perimeter in case of Round Bars only in a new column No. 5 in Table 37 (P 158)
37A	Indian Standard Rivets (Snap Head Rivets)	} New tables introduced
37B	Indian Standard Bolts and Nuts (Hexagon Head)	
45	Steel Unequal Angles	This table replaces Table 45 Unequal Angles (P 162) as it provides substantial additional data in eight more columns

TABLE 18A PSYCHROMETRIC PROPERTIES OF AIR*

NOTE — The values of enthalpy in columns 4, 5, 8 and 10 are measured from water at 0°C.

TEMPERATURE, °C	PROPERTIES OF WATER (OR ICE) AND STEAM					PROPERTIES OF DRY AIR AT A PRESSURE OF 76.00 cm Hg		PROPERTIES OF MIXTURE OF DRY AIR AND SATURATED STEAM AT A TOTAL PRESSURE OF 76.00 cm Hg		
	Saturation Pressure of Water and Steam		Enthalpy of Saturated Ice (-20° to 0°C)	Enthalpy of Saturated steam, kcal/kg	Specific Volume of Saturated Steam, m³/kg	Specific Volume, m³/kg	Enthalpy, kcal/kg	Volume of Mixture per kg of Dry Air, m³/kg	Enthalpy of Mixture per kg of Dry Air, kcal/kg	Specific Humidity of Mixture, g/kg
	cm Hg	10⁻³ gf/cm²	Enthalpy of Saturated Water (0° to 100°C), kcal/kg							10⁻³ kg/kg
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
-20	0.077 4	1.052	-89.39	588.5	1 132.4	0.716 5	-4.81	0.717 3	-4.433	0.633 5
-19	0.085 2	1.158	-88.89			0.719 8	-4.57	0.720 2	-4.153	0.697 3
-18	0.093 6	1.273	-88.42	589.4	1 862.0	0.722 2	-4.33	0.723 0	-3.873	0.766 9
-17	0.102 9	1.399	-87.94	589.8	788.3	0.725 1	-4.09	0.726 1	-3.588	0.842 8
-16	0.112 9	1.536	-87.50	590.3		0.727 9	-3.85	0.728 7	-3.306	0.918 2
-15	0.123 9	1.685	-87.00	590.7	721.0	0.730 8	-3.61	0.731 6	-3.006	1.015
-14	0.135 8	1.847	-86.54			0.733 6	-3.37	0.735 0	-2.706	1.113
-13	0.148 8	2.024	-86.06	591.2	605.3	0.736 5	-3.13	0.737 8	-2.402	1.220
-12	0.162 9	2.216	-85.56	592.0	554.9	0.739 3	-2.89	0.740 9	-2.092	1.336
-11	0.178 2	2.424	-85.11	592.5	509.2	0.742 2	-2.65	0.743 9	-1.776	1.462
-10	0.194 8	2.649	-84.61	592.9	467.5	0.744 9	-2.41	0.746 6	-1.456	1.597
-9	0.212 8	2.894	-84.11	593.3	429.6	0.747 8	-2.17	0.750 0	-1.126	1.746
-8	0.232 4	3.160	-83.61	593.8	395.0	0.750	-1.93	0.753 0	-0.792	1.907
-7	0.253 5	3.447	-83.13	594.2	363.4	0.753 3	-1.68	0.756 1	-0.444	2.081
-6	0.276 4	3.759	-82.67	594.7	334.6	0.756 3	-1.44	0.758 9	-0.093	2.269
-5	0.301 2	4.096	-82.17	595.1	308.1	0.759 2	-1.20	0.762 2	0.272	2.474
-4	0.327 9	4.459	-81.67	595.6	284.1	0.762 0	-0.96	0.765 6	0.642	2.695
-3	0.356 9	4.853	-81.17	596.0	262.1	0.764 9	-0.72	0.768 4	1.030	2.934
-2	0.388 2	5.279	-80.67	596.4	241.8	0.767 7	-0.48	0.771 7	1.422	3.192
-1	0.421 8	5.735	-80.17	596.8	223.3	0.770 5	-0.24	0.775 0	1.830	3.471
0	0.458 0	6.223	-79.67	597.2	206.3	0.773 4	0.00	0.777 8	2.256	3.771
1	0.492 4	6.694	0.00	597.6		0.776 2	0.24	0.781 2	2.666	4.057
2	0.529 1	7.19 3	1.01	598.0	192.7 180.0	0.779 0	0.48	0.784 5	3.090	4.362
3	0.568 1	7.72 3	2.01	598.5	168.2	0.781 9	0.72	0.787 9	3.526	4.686
4	0.609 7	8.289	3.02	599.0	157.3	0.784 7	0.96	0.791 3	3.978	5.031
5	0.653 9	8.890	4.04	599.4	147.2	0.787 5	1.20	0.794 7	4.439	5.400
6	0.701 0	9.530	5.03	599.8	137.8	0.790 4	1.44	0.798 0	4.919	5.793
7	0.751 0	10.210	6.03	600.2	129.1	0.793 2	1.69	0.801 4	5.411	6.210
8	0.804 1	10.932	7.03	600.7	121.0	0.796 0	1.93	0.804 8	5.918	6.654
9	0.860 5	11.699	8.04	601.1	113.4	0.798 9	2.17	0.808 2	6.449	7.127
10	0.920 4	12.513	9.04	601.6	106.4	0.8018	2.41	0.811 5	6.994	7.629
11	0.983 9	13.376	10.04	602.0	99.9	0.804 6	2.65	0.814 9	7.559	8.162
12	1.051 2	14.291	11.04	602.5	93.8	0.807 4	2.89	0.818 9	8.144	8.729
13	1.122 5	15.261	12.04	602.9	88.1	0.810 3	3.13	0.822 3	8.754	9.333
14	1.198 2	16.289	13.04	603.4	82.9	0.813 1	3.37	0.826 2	9.383	9.971
15	1.278 1	17.376	14.04	604.3	77.9	0.815 9	3.61	0.829 6	10.04	10.65
16	1.362 8	18.527	15.04	604.7	73.3	0.818 8	4.09 3.85	0.833 6	10.71	11.37
17	1.452 4	19.745	16.04	605.1	69.1	0.821 6		0.837 6	11.42	12.13
18	1.547	21.03	17.04	605.6	65.1	0.824 4	4.33	0.841 6	12.16	12.94
19	1.647	22.39	18.04	606.0	61.3	0.827 3	4.57	0.845 6	12.92	13.79

* Reproduced from ISI Handbook of quantities, conversion factors, formulae and tables.

(Continued)

TABLE 18A PSYCHROMETRIC PROPERTIES OF AIR—Contd

NOTE—The values of enthalpy in columns 4, 5, 8 and 10 are measured from water at 0°C.

TEMPERATURE, °C	PROPERTIES OF WATER (OR ICE) AND STEAM					PROPERTIES OF DRY AIR AT A PRESSURE OF 76.00 cm Hg		PROPERTIES OF MIXTURE OF DRY AIR AND SATURATED STEAM AT A TOTAL PRESSURE OF 76.00 cm Hg		
	Saturation Pressure of Water and Steam		Enthalpy of Saturated Ice (–20° to 0°C) Enthalpy of Saturated Water 0° to 100 °C), kcal/kg	Enthalpy of Saturated Steam, kcal/kg	Specific Volume of Saturated Steam, m³/kg	Specific Volume, m³/kg	Enthalpy kcal/kg	Volume of Mixture per kg of Dry Air, m³/kg	Enthalpy of Mixture per kg of Dry Air, kcal/kg	Specific Humidity per kg of Dry Air, 0–° kg/kg
	cm Hg	10 ^{–3} kgf/cm²								
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
20	1.753	23.83	20.03	606.0	57.84	0.830 1	4.81	10.849 6	13.72	14.70
21	1.864	25.34	21.03	606.5	54.56	0.833 0	5.05	0.854 1	14.55	15.66
22	1.982	26.94	22.03	606.9	51.49	0.835 8	5.29	0.858 0	15.41	16.68
23	2.106	28.63	23.02	607.3	48.63	0.838 7	5.53	0.862 6	16.32	17.76
24	2.237	30.41	24.02	607.8	45.94	0.8415	5.77	0.867 0	17.25	18.89
25	2.375	32.29	25.02	608.2	43.41	0.844 4	6.01	0.871 5	18.23	20.09
26	2.520	34.26	26.01	608.6	41.04	0.847 2	6.25	0.875 9	19.76	21.37
27	2.673	36.34	27.01	609.1	38.82	0.850 0	6.49	0.881 1	20.33	22.72
28	2.834	38.53	28.01	609.5	36.73	0.852 9	6.73	0.885 8	21.45	24.14
29	3.003	40.83	29.00	610.0	34.77	0.855 7	6.97	0.890 8	22.62	25.65
30	3.181	43.25	30.00	610.4	32.93	0.858 5	7.21	0.895 8	23.83	27.23
31	3.369	45.80	31.00	610.8	31.20	0.861 3	7.45	0.901 4	25.11	28.91
32	3.565	48.47	32.00	611.3	29.58	0.864 2	7.69	0.907 0	26.45	30.69
33	3.772	51.28	32.99	611.7	28.05	0.867 1	7.94	0.912 7	27.85	32.56
34	3.989	54.23	33.99	612.1	26.61	0.869 9	8.18	0.918 3	29.32	34.53
35	4.217	57.33	34.99	612.5	25.25	0.872 8	8.42	0.923 9	30.85	36.63
36	4.455	60.57	35.99	613.0	23.97	0.875 6	8.66	0.930 1	32.46	38.83
37	4.706	63.98	36.98	613.4	22.77	0.878 4	8.90	0.936 4	34.15	41.17
38	4.969	67.55	37.98	613.9	21.63	0.881 3	9.14	0.942 9	35.92	43.63
39	5.244	71.29	38.98	614.3	20.56	0.884 1	9.38	0.949 6	37.78	46.26
40	5.531	75.20	39.98	614.7	19.55	0.886 9	9.62	0.956 3	39.72	48.97
41	5.833	79.30	40.98	615.2	18.60	0.889 7	9.86	0.963 7	41.77	51.87
42	6.149	83.60	41.97	615.6	17.70	0.892 6	10.10	0.971 5	43.92	54.93
43	6.480	88.09	42.97	616.0	16.85	0.895 4	10.34	0.978 7	46.17	58.17
44	6.825	92.79	43.97	616.4	16.04	0.898 3	10.58	0.987 2	48.54	61.58
45	7.187	97.71	44.96	616.8	15.28	0.901 1	10.82	0.995 1	51.03	65.19
46	7.564	102.84	45.96	617.2	14.56	0.904 0	11.06	1.000 0	53.66	69.01
47	7.959	108.21	46.96	617.7	13.88	0.906 8	11.30	1.013 0	56.42	73.04
48	8.372	113.82	47.95	618.1	13.24	0.909 7	11.54	1.022 3	59.33	77.31
49	8.802	119.67	48.95	618.5	12.63	0.912 5	11.78	1.031 8	62.39	81.81
50	9.252	125.78	49.95	619.0	12.05	0.915 3	12.02	1.042 5	65.61	86.57
51	9.754	132.16	50.95	619.4	11.50	0.918 2	12.27	1.053 1	69.02	91.61
52	10.210	138.81	51.95	619.8	10.98	0.921 0	12.51	1.063 7	72.61	96.95
53	10.721	145.75	52.94	620.2	10.49	0.923 8	12.75	1.075 6	76.39	102.6
54	11.253	152.98	53.94	620.6	10.02	0.926 6	12.99	1.087 7	80.39	108.6
55	11.806	160.51	54.94	621.0	9.584	0.929 5	13.23	1.100 6	84.62	114.9
56	12.383	168.35	55.94	621.5	9.164	0.932 3	13.47	1.114 1	89.10	121.7
57	12.985	176.53	56.94	621.9	8.764	0.935 2	13.71	1.127 9	93.84	128.8
58	13.611	185.04	57.94	622.3	8.385	0.938 0	13.95	1.142 8	98.86	136.4
59	14.262	193.90	58.94	622.7	8.025	0.940 9	14.19	1.158 5	104.2	144.5

(Continued)

TABLE 18A PSYCHROMETRIC PROPERTIES OF AIR—Contd

Note—The values of enthalpy in columns 4, 5, 8 and 10 are measured from water at 0%.

TEMPERATURE, °C	PROPERTIES OF WATER (OR ICE) AND STEAM				PROPERTIES OF DRY AIR AT A PRESSURE OF 76.00 cm Hg		PROPERTIES OF MIXTURE OF DRY AIR AND SATURATED STEAM AT A TOTAL PRESSURE OF 76.00 cm Hg			
	Saturation Pressure of Water and Steam		Enthalpy of Saturated Ice (−20° to 0°C) Enthalpy of Saturated Water (0° to 100°C), kcal/kg	Enthalpy of Saturated Steam, kcal/kg	Specific Volume of Saturated Steam, m³/kg	Specific Volume, m³/kg	Enthalpy, kcal/kg	Volume of Mixture per kg of Dry Air, m³/kg	Enthalpy of Mixture per kg of Dry Air, kcal/kg	Specific Humidity per kg of Dry Air, g/kg
	cm Hg	10⁻³ kgf/cm²								
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
60	14.94	203.1	59.94	623.2	7.682	0.943 7	1443	1.174 8	109.8	153.0
61	15.65	212.7	60.94	623.6	7.356	0.946 5	1467	1.192 2	115.8	162.2
62	16.38	222.7	61.94	624.0	7.046	0.949 4	14.91	1.210 2	122.2	171.9
63	17.14	233.0	62.94	624.4	6.752	0.952 2	15.15	1.229 5	128.9	182.3
64	17.93	243.8	63.94	624.8	6.473	0.955 0	15.39	1.250 3	136.2	193.3
65	18.76	255.0	64.93	625.2	6.206	0.957 9	15.63	1.271 6	143.9	205.1
66	19.61	266.6	65.93	625.6	5.951	0.960 7	15.87	1.295 2	152.2	217.9
67	20.50	278.7	66.93	626.0	5.709	0.963 7	16.12	1.319 8	161.0	231.3
68	21.42	291.2	67.93	626.4	5.478	0.966 4	16.36	1.345 9	170.4	245.9
69	22.38	304.2	68.93	626.9	5.258	0.969 2	16.60	1.374 1	180.6	261.5
70	23.37	317.7	69.93	627.3	5.048	0.972 1	16.84	1.404 0	191.4	278.3
71	24.40	331.7	70.93	627.7	4.848	0.974 9	17.08	1.436 4	203.2	296.5
72	25.47	346.3	71.93	628.1	4.658	0.977 8	17.32	1.470 9	215.8	316.1
73	26.58	361.3	72.93	628.5	4.476	0.980 6	17.56	1.508 2	229.5	337.2
74	27.72	376.9	73.94	628.9	4.302	0.983 4	17.80	1.548 5	244.4	360.3
75	28.91	393.1	74.94	629.3	4.136	0.986 3	18.04	1.5919	260.5	385.1
76	30.14	409.8	75.94	629.7	3.977	0.989 1	18.28	1.639 6	278.1	412.6
77	31.42	427.2	76.94	630.1	3.826	0.991 9	18.52	1.691 7	297.3	442.4
78	32.74	445.1	77.95	630.5	3.681	0.994 8	18.76	1.748 1	318.4	475.2
79	34.11	463.7	78.95	630.9	3.542	0.997 6	19.01	1.810 2	341.6	511.3
80	35.52	482.9	79.95	631.3	3.410	1.000 4	19.25	1.878 4	367.2	551.1
81	36.98	502.8	80.95	631.7	3.283	1.003 3	19.49	1.955 1	395.8	595.6
82	38.50	523.4	81.95	632.1	3.162	1.006 2	19.73	2.040 0	427.7	645.3
83	40.07	544.7	82.96	632.5	3.047	1.009 0	19.97	2.134 6	463.4	701.1
84	41.68	566.7	83.96	632.8	2.936	1.011 8	20.21	2.241 6	503.7	763.9
85	43.35	589.4	84.96	633.2	2.830	1.014 7	20.46	2.362 8	549.4	835.3
86	45.08	612.9	85.96	633.6	2.728	1.017 5	20.70	2.502 1	602.3	917.7
87	46.87	637.2	86.97	634.0	2.630	1.020 3	20.94	2.663 6	663.4	1013
88	48.72	662.3	87.97	634.4	2.537	1.023 1	21.18	2.851 4	734.7	1125
89	50.62	688.2	88.98	634.7	2.447	1.026 0	21.42	3.073 7	819.1	1256
90	52.59	714.9	89.98	635.1	2.361	1.028 8	21.66	3.339 8	920.2	1414
91	54.62	742.5	90.98	635.5	2.275	—	—	—	—	—
92	56.71	771.0	91.99	635.9	2.200	—	—	—	—	—
93	58.87	800.4	93.00	636.3	2.124	—	—	—	—	—
94	61.10	830.7	94.00	636.7	2.051	—	—	—	—	—
95	63.40	861.9	95.01	637.0	1.981	—	—	—	—	—
96	65.77	894.2	96.01	637.4	1.914	—	—	—	—	—
97	68.22	927.4	97.02	637.8	1.851	—	—	—	—	—
98	70.73	961.6	98.02	638.2	1.786	—	—	—	—	—
99	73.33	996.9	99.03	638.5	1.730	—	—	—	—	—
100	76.00	1033.2	100.04	638.9	1.675	—	—	—	—	—

TABLE 25A SOME PHYSICAL CONSTANTS*

SYMBOL	QUANTITY	VALUE	log ₁₀ (VALUE)
$a_0 = \frac{h^2}{4\pi^2 m e^2}$	First Bohr Radius	$5.291\,73 \times 10^{-9}$ cm	9.723 60
$\alpha = \frac{2\pi e^2}{hc}$	Fine Structure Constant	$7.297\,29 \times 10^{-3}$	3.863 16
α^{-1}		$1.370\,372 \times 10^3$	2.136 85
c	Velocity of Light	$2.997\,929 \times 10^{10}$ cm/s	10.476 82
$c_1 = 8\pi hc$	First Radiation constant	$4.991\,9 \times 10^{-15}$ erg cm	15.698 26
$c_2 = hc/k$	Second Radiation Constant	1.438 80 cm deg	0.158 00
e	Electronic charge	$4.802\,90 \times 10^{-10}$ esu	10.681 50
e/c		$1.602\,08 \times 10^{-20}$ emu	20.204 69
e/m	Charge to Mass Ratio of Electron	$5.273\,02 \times 10^{17}$ esu/g	17.722 06
e/mc		$1.758\,89 \times 10^7$ emu/g	7.245 25
$F = Ne/c$	Faraday Constant	$9.652\,1 \times 10^9$ emu (g-mole) ⁻¹	3.984 63
G	Gravitational Constant	6.670×10^{-8} g ⁻¹ cm ³ s ⁻²	8.824 13
h	Planck's constant	$6.625\,3 \times 10^{-27}$ erg s	27.821 21
$\hbar = h/2\pi$		$1.054\,45 \times 10^{-27}$ erg s	27.023 02
h/e		$1.379\,43 \times 10^{-16}$ s/esu	17.139 69
k	Boltzmann's Constant	$1.380\,47 \times 10^{-16}$ erg/deg	16.140 03
$\lambda_{ce} = h/mc$	Compton Wavelength of Electron	$2.426\,27 \times 10^{-10}$ cm	10.384 94
$\lambda_{cp} = h/m_p c$	Compton Wavelength of Proton	1.32141×10^{-13} cm	13.121 03
$\lambda_{cn} = h/m_n c$	Compton Wavelength of Neutron	$1.319\,59 \times 10^{-13}$ cm	13.120 44
$\lambda_{\max} T = b$	Wien-displacement-law Constant	0.289 782 cm deg	1.462 07
M_e	Atomic Mass of Electron	$5.487\,62 \times 10^{-4}$	4.739 38
M_H	Atomic Mass of Hydrogen	1.008 144	0.003 52
M_n	Atomic Mass of Neutron	1.008 983	0.003 89
M_p	Atomic Mass of Proton	1.007 595	0.003 29
M_p/M_e	Ratio of Proton to Electron Mass	$1.836\,12 \times 10^3$	3.263 89
m	Electron Rest Mass	$9.108\,5 \times 10^{-28}$ g	28.959 44
$m_n = M_n/N$	Neutron Rest Mass	$1.674\,73 \times 10^{-24}$ g	24.223 95
$m_p = M_p/N$	Proton Rest Mass	$1.672\,42 \times 10^{-24}$ g	24.223 34
$\mu = m M_p/M_H$	Reduced Mass of Electron in Hydrogen Atom	$9.103\,5 \times 10^{-28}$ g	28.959 20
μ_e	Magnetic Moment of Electron	$9.283\,9 \times 10^{-21}$ erg/gauss	21.967 73
$\mu_n = he/4\pi m_p c$	Nuclear Magnetron	$5.050\,4 \times 10^{-24}$ erg/gauss	24.703 33
$\mu_p = he/4\pi m c$	Bohr Magnetron	$9.273\,2 \times 10^{-21}$ erg/gauss	21.967 23
N	Avogadro's Number	$6.024\,8 \times 10^{23}$ mole ⁻¹	23.779 94
n_0	Loschmidt's Number	$2.687\,09 \times 10^{19}$ cm ⁻³	19.429 29
$R_0 = Nk$	Ideal Gas Constant	$8.317\,0 \times 10^7$ erg/mole deg	7.919 97
$R_H = \frac{R_\infty}{1 + \frac{M_p}{M_H}}$	Rydberg Constant for Hydrogen	$1.096\,78 \times 10^8$ cm ⁻¹	5.040 13
$R_\infty = \frac{2\pi^2 m e^4}{ch^3}$	Rydberg Constant for Infinite Mass	$1.097\,37 \times 10^8$ cm ⁻¹	5.040 36
r	Ratio of Physical to Chemical Scale of Atomic Weight	1.000 275	0.000 11
$r_0 = e^2/mc^2$	Classical Electron Radius	$2.817\,85 \times 10^{-13}$ cm	13.449 92
$\sigma = \frac{2\pi^5 k^4}{15c^2 h^3}$	Stefan-Boltzmann Constant	$5.669\,2 \times 10^{-5}$ erg/cm ² s deg ⁴	5.753 53
$V_0 = \frac{N}{n_0}$	Molar Volume	$2.242\,1 \times 10^4$ cm ³ /mole	4.350 66

*Reproduced from IS1 Handbook of quantities, conversion factors, formulae and tables.

TABLE 32A ROLLED STEEL HEAVY BEAMS

DESIGNATION (1)	WEIGHT PER METRE w (2) kg	SECTIONAL AREA a (3) cm ²	DEPTH OF SECTION h (4) mm	WIDTH OF FLANGE b (5) mm	THICK- NESS OF FLANGE t_f (6) mm	THICK- NESS OF WEB A (7) mm	Momma of INERTIA		RADI OF GYRATION		MODULI OF SECTION	
							I_{xx}	I_{yy}	r_{xx}	r_{yy}	Z_{xx}	Z_{yy}
							(8) cm ⁴	(9) cm ⁴	(10) cm	(11) cm	(12) cm ³	(13) cm ³
ISHB 150	27.1	3448	150	150	9.0	5.4	1 455.6	431.7	6.50	3.54	194.1	57.6
ISHB 150	30.6	38.98	150	150	9.0	8.4	1540.0	460.3	6.29	3.44	205.3	60.2
ISHB 150	34.6	44.08	150	150	9.0	11.8	1 635.6	494.9	6.09	3.35	218.1	63.2
ISHB 200	37.3	47.54	200	200	9.0	6.1	3 608.4	967.1	8.71	4.51	360.8	96.7
ISHB 200	40.0	50.94	200	200	9.0	7.8	3 721.8	994.6	8.55	4.42	372.2	98.6
ISHB 225	43.1	54.94	225	225	9.1	6.5	5 279.5	1 353.8	9.80	4.96	469.3	120.3
ISHB 225	46.8	59.66	225	225	9.1	8.6	5 478.8	1 396.6	9.58	4.84	487.0	123.0
ISHB 250	51.0	64.96	250	250	9.7	6.9	7 736.5	1 961.3	10.91	5.49	618.9	156.9
ISHB 250	54.7	69.71	250	250	9.7	8.8	7 983.9	2 011.7	10.70	5.37	638.7	159.7
ISHB 300	58.8	74.85	300	250	10.6	7.6	12 545.2	2 193.6	12.95	5.41	836.3	175.5
ISHB 300	63.0	80.25	300	250	10.6	9.4	12 950.2	2 246.7	12.70	5.29	863.3	178.4
ISHB 350	67.4	85.91	350	250	11.6	8.3	19 159.7	2 451.4	14.93	5.34	1094.8	196.1
ISHB 350	72.4	92.21	350	250	11.6	10.1	19 802.8	2 510.5	14.65	5.22	1 131.6	199.4
ISHB 400	77.4	98.66	400	250	12.7	9.1	28 083.5	2 728.3	16.87	5.26	1 404.2	218.3
ISHB 400	82.2	104.66	400	250	12.7	10.6	28 823.5	2 783.0	16.61	5.16	1 444.2	221.3
ISHB 450	87.2	111.14	450	250	13.7	9.8	39 210.8	2 985.2	18.78	5.18	1 742.7	238.8
ISHB 450	92.5	117.89	450	250	13.7	11.3	40 349.9	3 045.0	18.50	5.08	1 793.3	242.1

TABLE 33 ROLLED STEEL CHANNELS

DESIGNATION (1)	CENTRE OF GRAVITY C_{yy} (12) cm	DESIGNATION (1)	CENTRE OF GRAVITY C_{yy} (12) cm
ISJC 100	1.40	ISMC 75	1.32
ISJC 125	1.64	ISMC 100	1.54
ISJC 150	1.67	ISMC 125	1.96
ISJC 175	1.75	ISMC 125	1.92
ISJC 200	1.97	ISMC 150	2.23
ISLC 75	1.35	ISMC 150	2.17
ISLC 100	1.62	ISMC 175	2.21
	2.04	ISMC 175	2.14
ISLC (25	1.87	ISMC 200	2.18
ISLC (P) 125	2.39	ISMC 200	2.11
ISLC 50			
ISLC (P) 150	2.24	ISMC 225	2.31
ISLC 175	2.40	ISMC 225	2.22
ISLC 200	2.36	ISMC 250	2.30
ISLC (P) 200	2.23	ISMC 250	2.23
ISLC 225	2.47	ISMC 250	2.19
ISLC 250	2.71	ISMC 300	2.37
ISLC 300	2.56	ISMC 300	2.26
ISLC (P) 300	2.32	ISMC 300	2.22
ISLC 350	2.42	ISMC 350	2.46
ISLC 400	2.37	ISMC 400	2.43

TABLE 34 STEEL EQUAL ANGLES

DESIGNATION	CENTRE OF GRAVITY $C_{xx}=C_{yy}$	DESIGNATION	CENTRE OF GRAVITY $C_{xx}=C_{yy}$
(1)	(11)	(1)	(11)
	cm		cm
ISA 2020	0.59 0.63	ISA 7070	1.89 1.94 2.02 2.10
ISA 2525	0.71 0.75 0.79	ISA 7575	2.02 2.06 2.14 2.22
ISA 3030	0.83 0.87 0.92	ISA 8080	2.18 2.27 2.34 2.42
ISA 3535	0.95 1.00 1.04 1.08	ISA 9090	2.42 2.51 2.59 2.66
ISA 4040	1.08 1.12 1.16 1.20	ISA 100100	2.67 2.76 2.84 2.92
ISA 4545	1.20 1.25 1.29 1.33	ISA 110110	3.00 3.09 3.17 3.32
ISA 5050	1.32 1.37 1.41 1.45	ISA 130130	3.50 3.59 3.67 3.82
ISA 5555	1.53 1.57 1.65 1.72	ISA 150150	4.08 4.16 4.31 4.46
ISA 6060	1.65 1.69 1.77 1.85	ISA 200200	5.39 5.56 5.71 5.90
ISA 6565	1.77 1.81 1.89 1.97		

TABLE 55 STEEL TEE BAR8

DESIGNATION	CENTRE OF GRAVITY POSITION C_{xx}	DESIGNATION	CENTRE OF GRAVITY POSITION C_{xx}
(1)	(10)	(1)	(10)
	cm		cm
ISNT 20		ISNT 60	1.56
ISNT 30	0.60 0.82	ISNT 75	2.04
ISNT 40	1.14	ISNT 100	2.62
ISNT 50	1.35	ISNT 150	3.61

TABLE 37 ROUND AND SQUARE STEEL BARS

Round Bars

DESIGNATION (1)	PERIMETER (5) c m	DESIGNATION (1)	PERIMETER (5) cm
ISRO 5	1.88	ISRO 45	14.1
ISRO 6	2.51	ISRO 50	15.7
ISRO 8		ISRO 56	17.6
ISRO 10	3.14	ISRO 63	19.8
ISRO 12	3.77	ISRO 71	22.3
ISRO 14	4.39	ISRO 80	25.1
ISRO 18	5.02	ISRO 90	28.3
ISRO 20	5.65	ISRO 100	31.4
ISRO 22	6.28	ISRO 110	34.6
ISRO 25	6.91	ISRO 125	39.3
ISRO 28	7.85	ISRO 140	44.0
ISRO 32	8.79	ISRO 160	50.3
ISRO 36	10.1	ISRO 180	56.6
ISRO 40	11.3	ISRO 200	62.8
	12.6		

TABLE 37A INDIAN STANDARD RIVETS (SNAP HEAD RIVETS)

[Calculated weight of rivets in kg/l 000 pieces (based on 7.85 kg/dm³)]

NOMINAL DIAMETER OF RIVETS, → D, mm LENGTH, l, mm ↓	10	12	16	20	22	24
	Weight of rivets in kg/l 000 pieces ~					
20	18.5	—	—	—	—	—
22	20.9 19.7	—	—	—	—	—
24	—	—	—	—	—	—
26	23.4 22.2	—	—	—	—	—
28	24.7	37.6	—	—	—	—
30	—	—	—	—	—	—
31.5	25.9	40.7	80	—	—	—
35	30	—	—	—	—	—
35.5	—	44.3	86.3	—	—	—
40	32.1	48.3	93.4	—	—	—
45	35.1	52.7	101	168	—	—
55	41.3 38.2	57.2	109	180	228	—
56	—	62.5	119	195	258	301
63	44.4	68.7	130 125	212 205	266	326 315
65	47.5	—	—	230	—	—
70	50.6	75.8 74.9	142 141	232	287 290	354 351
80	—	83.8	157	254	317	386
85	—	88.3	164	279	347	404
98	—	—	172	302	362	422
100	—	—	188	302	362	440
106	—	—	198	319	377	457
112	—	—	207	348	395	479
118	—	—	—	365	413	500
125	—	—	—	—	431	521
132	—	—	—	—	473 452	546
140	—	—	—	—	—	571
150	—	—	—	—	497	600
160	—	—	—	—	—	635
						671
Weight per 1000 heads in kg ~	7.38	12.77	30.25	59.04	78.6	102.1
For each additional 10 mm length of shank add 1 000 pieces, kg ~ per	6.17	8.88	15.8	25.0	29.88	35.54

TABLE 37B INDIAN STANDARD BOLTS AND NUTS (HEXAGON HEAD)

[Calculated weight in kg/1000 pieces of bolts and nub (based on 7.85 kg/dm³)]

NOMINAL SIZE, d, mm→	M6	M8	M10	M12	M14	M16	M18	M20	M22	M24	M27	M30	M33	M36	M39	M42	M45	M48	M52
	Weight of bolts in kg/1 000 pieces≈																		
LENGTH, l, mm ↓																			
16	7.43	15.9	32.1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
20	8.12	17.1	34.1	48.9	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
25	8.97	18.7	36.6	52.5	66.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—
30	10.1	20.7	39.1	56.1	72.6	107	136	186	212	—	—	—	—	—	—	—	—	—	—
35	11.2	22.7	42.2	59.7	78.9	114	146	196	227	—	—	—	—	—	—	—	—	—	—
40	12.3	24.7	45.3	64.1	85.2	121	157	207	244	347	—	—	—	—	—	—	—	—	—
45	13.4	26.7	48.4	68.5	91.4	128	167	217	260	362	—	—	—	—	—	—	—	—	—
50	14.5	28.7	51.5	72.9	97.7	136	177	227	277	377	—	—	—	—	—	—	—	—	—
55	15.6	30.7	54.6	77.3	104	144	188	238	293	392	—	—	—	—	—	—	—	—	—
60	16.7	32.7	57.7	81.7	110	152	198	251	309	407	570	—	—	—	—	—	—	—	—
65	17.8	34.7	60.8	86.1	116	160	208	264	326	422	589	—	—	—	—	—	—	—	—
70	18.9	36.7	63.9	90.5	123	168	219	277	343	440	608	—	—	—	—	—	—	—	—
75	20.0	38.7	67.0	95	129	176	229	290	359	458	630	—	—	—	—	—	—	—	—
80	21.1	40.7	70.1	100	135	182	239	303	376	476	653	853	1080	—	—	—	—	—	—
90	23.3	44.7	76.3	109	148	198	260	329	407	511	698	908	1 150	—	—	—	—	—	—
100	25.5	48.7	82.5	118	160	214	280	355	436	547	743	963	1220	1520	1870	2 230	2660	—	—
110	—	52.7	88.7	127	172	230	301	381	469	583	788	1018	1285	1600	1965	2340	2 785	3 300	—
120	—	56.7	94.9	136	185	246	322	407	500	618	833	1070	1 350	1680	2060	2450	2 910	3 440	4 140
130	—	—	101	145	197	262	342	433	531	654	878	1 125	1415	1 750	2 155	2 560	3 035	3580	4 305
140	—	—	107	154	210	278	363	459	562	689	923	1180	1480	1 840	2250	2 670	3160	3 720	4 470
150	—	—	113	163	222	294	384	485	593	725	968	1230	1 550	1920	2 340	2780	3280	3860	4 635
160	—	—	119	172	235	310	404	501	624	760	1010	1290	1620	2000	2430	2890	3400	4000	4800
170	—	—	125	181	247	326	425	527	655	796	1055	1345	1685	2080	2 525	3000	3 525	4 140	4 935
180	—	—	131	190	260	342	445	553	686	831	1100	1 400	1 750	2 160	2 620	3 110	3650	4280	5130
190	—	—	137	199	272.0	358	466	579	717	867	1 145	1455	1815	2240	2 715	3 220	3 775	4 420	5 295
200	—	—	143	208	284	374	487	605	748	902	1190	1510	1880	2 320	2 810	3 330	3 900	4560	5 460
Weight of nuts in kg per 1 000 pieces≈	2.32	4.82	10.9	15.9	22.4	30.8	43.2	60.3	80.3	103	154	216	271	369	472	610	750	924	1 130
Weight of 10 mm of shank in kg per 1000 pieces≈	2.2	4	6	9	12.4	16	20.7	25	31.2	35.5	44.7	54.7	66	80	95	110	125	140	165

TABLE 45 STEEL UNEQUAL ANGLES

DESIGNATION	SIZE $A \times B$	THICKNESS t	SECTIONAL AREA a	WEIGHT PER METRE w	MOMENTS OF INERTIA			MODULI OF SECTION		CENTRE OF GRAVITY	
					I_{xx}	I_{yy} (Max)	I_{zz}	Z_{xx}	Z_{yy}	C_{xx}	C_{yy}
(1)	(2) mm x mm	(3) mm	(4) mm ²	(5) kg	(6) cm ⁴	(7) cm	(8) cm ⁴	(9) cm ³	(10) cm ³	(11) cm	(12) cm
ISA 3020	30x20	3.0	1.41	1.1	1.2	1.4	0.4	0.6	0.3	0.98	0.49
		4.0	1.84	1.4	1.5	1.8	0.5	0.8	0.4	1.02	0.53
		5.0	2.25	1.8	1.9	2.1	0.6	1.0	0.4	1.06	0.57
ISA 4025	40x25	3.0	1.88	1.5	3.0	3.3	0.9	1.1	0.5	1.30	0.57
		4.0	2.46	1.9	3.8	4.3	1.1	1.4	0.6	1.35	0.62
		5.0	3.02	2.4	4.6	5.1	1.4	1.8	0.7	1.39	0.66
		6.0	3.56	2.8	5.4	5.9	1.6	2.1	0.9	1.43	0.69
ISA, 4530	45x30	3.0	2.18	1.7	4.4	5.0	1.5	1.4	0.7	1.42	0.69
		4.0	2.86	2.2	5.7	6.5	2.0	1.9	0.9	1.47	0.73
		5.0	3.52	2.8	6.9	7.9	2.4	2.3	1.1	1.51	0.77
		6.0	4.16	3.3	8.0	9.2	2.8	2.7	1.3	1.55	0.81
ISA 5030	50x30	3.0	2.34	1.8	5.9	6.5	1.6	1.7	0.7	1.63	0.65
		4.0	3.07	2.4	7.7	8.5	2.1	2.3	0.9	1.68	0.70
		5.0	3.78	3.0	9.3	10.3	2.5	2.8	1.1	1.72	0.74
		6.0	4.47	3.5	10.9	11.9	2.9	3.4	1.3	1.76	0.78
ISA 6040	60x40	5.0	4.76	3.7	16.9	19.5	6.0	4.2	2.0	1.95	0.96
		6.0	5.65	4.4	19.9	22.8	7.0	5.0	2.3	1.99	1.00
		8.0	7.37	5.8	25.4	29.0	9.8	6.5	3.0	2.07	1.08
ISA 6545	65x45	5.0	5.26	4.1	22.1	25.9	8.6	5.0	2.5	2.07	1.08
		6.0	6.25	4.9	26.0	30.4	10.1	5.9	3.0	2.11	1.12
		8.0	8.17	6.4	33.2	38.7	12.8	7.7	3.9	2.19	1.20
ISA 7045	70x45	5.0	5.52	4.3	27.2	30.9	8.8	5.7	2.5	2.27	1.04
		6.0	6.56	5.2	32.0	36.3	10.3	6.8	3.0	2.32	1.09
		8.0	8.58	6.7	41.0	46.3	13.1	8.9	3.9	2.40	1.16
		10.0	10.52	8.3	49.3	55.4	15.6	10.9	4.8	2.48	1.24
ISA 7550	75x50	5.0	6.02	4.7	34.1	39.4	12.2	6.7	3.2	2.39	1.16
		6.0	7.16	5.6	40.3	46.4	14.3	8.0	3.8	2.44	1.20
		8.0	9.38	7.4	51.8	59.4	18.3	10.4	4.9	2.52	1.28
		10.0	11.52	9.0	62.3	71.2	21.8	12.7	6.0	2.60	1.36
ISA 8050	80x50	5.0	6.27	4.9	40.6	45.7	12.3	7.5	3.2	2.60	1.12
		6.0	7.46	5.9	48.0	53.9	14.4	9.0	3.8	2.64	1.16
		8.0	9.78	7.7	61.9	69.3	18.5	11.7	4.9	2.73	1.24
		10.0	12.02	9.4	74.7	83.3	22.1	14.4	6.0	2.81	1.32
ISA 9060	90x60	6.0	8.65	6.8	70.6	81.5	25.2	11.5	5.5	2.87	1.39
		8.0	11.37	8.9	91.5	105.3	32.4	15.1	7.2	2.96	1.48
		10.0	14.01	11.0	110.9	127.3	39.1	18.6	8.8	3.04	1.55
		12.0	16.57	13.0	129.1	147.5	45.2	22.0	10.3	3.12	1.63
ISA 10065	100x65	6.0	9.55	7.5	96.7	110.6	32.4	14.2	6.4	3.19	1.47
		8.0	12.57	9.9	125.9	143.6	41.9	18.7	8.5	3.28	1.55
		10.0	15.51	12.2	153.2	174.2	50.7	23.1	10.4	3.37	1.63
ISA 10075	100x75	6.0	10.14	8.0	100.9	124.0	48.7	14.4	8.5	3.01	1.78
		8.0	13.36	10.5	131.6	161.3	63.3	19.1	11.2	3.10	1.87
		10.0	16.50	13.0	160.4	196.1	76.9	23.6	13.8	3.19	1.95
		12.0	19.56	15.4	187.5	228.4	89.5	27.9	16.3	3.27	2.03
ISA 12575	125x75	6.0	11.66	9.2	187.8	208.9	51.6	22.2	8.7	4.05	1.59
		8.0	15.38	12.1	245.5	272.8	67.2	29.4	11.5	4.15	1.68
		10.0	19.02	14.9	300.3	332.9	81.6	36.3	14.2	4.24	1.76
ISA 12595	125x95	6.0	12.92	10.1	205.5	254.0	103.6	23.4	14.3	3.72	2.24
		8.0	17.04	13.4	268.3	331.4	134.7	30.9	18.8	3.80	2.32
		10.0	21.08	16.5	328.0	404.1	164.1	38.1	23.1	3.89	2.40
		12.0	25.04	19.7	384.8	473.6	191.8	45.1	27.3	3.97	2.48
ISA 15075	150 x 75	8.0	17.48	13.7	410.3	435.7	71.1	42.0	11.9	5.24	1.54
		10.0	21.62	17.0	502.2	532.7	86.3	51.9	14.7	5.33	1.62
		12.0	25.68	20.0	590.0	625.1	100.4	61.6	17.3	5.42	1.70

(Continued)

TABLE 45 STEEL UNEQUAL ANGLES — *Contd.*

DESIGNA- TION	SIZE $A \times B$	THICK- NESS t	SEC- TIONAL AREA a	WEIGHT PER METRE w	MOMENTS OF INERTIA			MODULI OF SECTION		CENTRE OF GRAVITY	
					I_{xx}	I_{yy} (I_{zz})	I_{yy}	Z_{xx}	Z_{yy}	C_{xx}	C_{yy}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	mm $\times$ mm	mm	m ²	kg	cm ⁴	cm ⁴	cm ⁴	cm ³	cm ³	cm	cm
ISA 150115	150x115	8.0	20.72			589.6	244.4	45.1	28.0	4.48	2.76
		10.0	25.66	20.1 16.9	474.4 501.8	722.6	298.8	55.8	34.5	4.57	2.84
		12.0	40.00	24.0	684.8	849.5	350.6	66.2	40.8	9.65	2.92
		10.0		31.4	878.5	1 085.9	446.8	86.2	53.0	4.81	3.07
ISA 200100	200x100		29.21	22.9	1 227.8	1 304.9	214.7	94.3		6.98	2.03
		12.0	34.77	27.3	1 449.2	1 539.0	251.5	112.1	91.9 26.9	7.07	2.11
		16.0	45.65	35.8	1 870.1	1 982.1	319.6	146.5	41.3	7.23	2.27
		10.0									
ISA 200150	200x150		34.29	26.9	1 409.2	1 729.6	688.9	100.8	60.2	6.02	3.55
		12.0	40.85	32.1	1 665.6	2 043.3	812.1	119.9	71.4	6.11	3.63
		16.0	53.73	42.2	2 155.2	2 638.6	1 044.9	157.0	93.2	6.27	3.79
		20.0	66.29	52.0	2 615.1	3 192.7	1 260.8	192.6	114.0	6.42	3.94